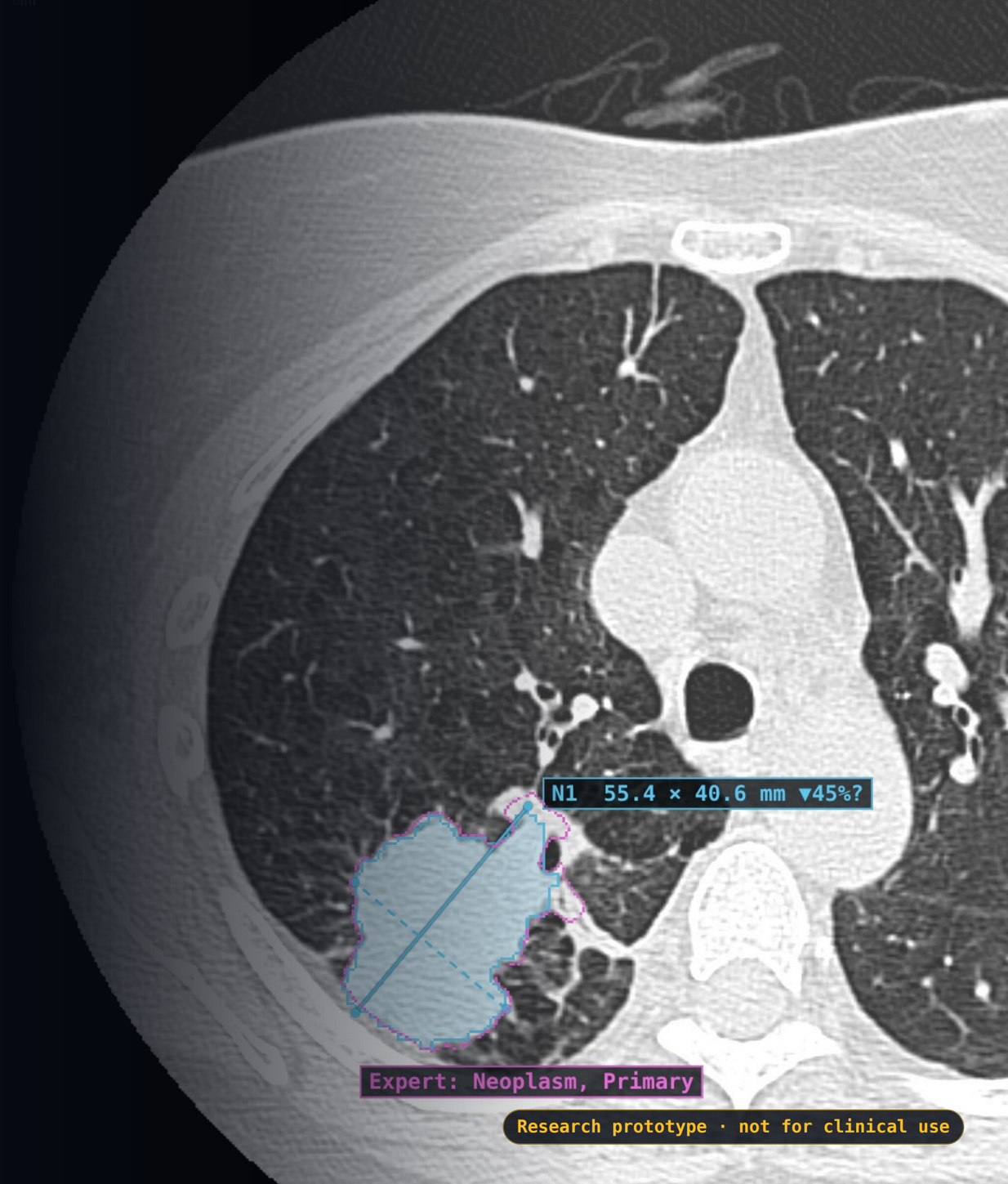
GUIDED WALKTHROUGH

Nodule Research Copilot

Faster CT nodule measurement for research.
The AI does the first pass. The researcher stays in control.



PRESENTED BY Yuval · Forward Deployed Engineer assignment, Base44

FOR Dr. Daniel Reyes & Dr. Naomi Hart · Thoracic Oncology Research Lab

DATE October 2026

Halve the time per scan, without losing quality

TODAY · DANIEL'S ESTIMATE

25–40 min

per baseline scan, mostly spent **finding** nodules slice by slice and **outlining** them for volume

→

GOAL · NAOMI'S TARGET

12–20 min

about **half the time**, with the same quality of measurement

Accurate

Finds and measures nodules close to an expert's outline, and shows where it fails.

In control

Every number is reviewed, correctable, and approved by the researcher.

Faster, measured

Time saved is measured on real work, not promised.

Times: Daniel's own estimate from the discovery call, not measured by us

From the hospital scan to your sheet

1

Export the scan

From the hospital system to a folder on the lab computer, as for any research scan.

YOU · ONCE PER SCAN

2

AI runs

One command for the whole folder. Can run overnight.

MACHINE · ~97 S PER SCAN

3

Open in Chrome

All scans of the patient, AI candidates already loaded.

MACHINE · SECONDS

4

Review & approve

Accept or reject, fix edges, add what was missed.

YOU · YOUR JUDGEMENT

5

CSV in your columns

Approved nodules export in your sheet's columns, ready for Excel. No re-typing.

AUTOMATIC

Scans stay on the computer

The browser blocks other sites (verified). The AI's usage statistics are switched off by our run script.

One install, no server

One Python package on the lab computer. No server, no connection to the hospital system.

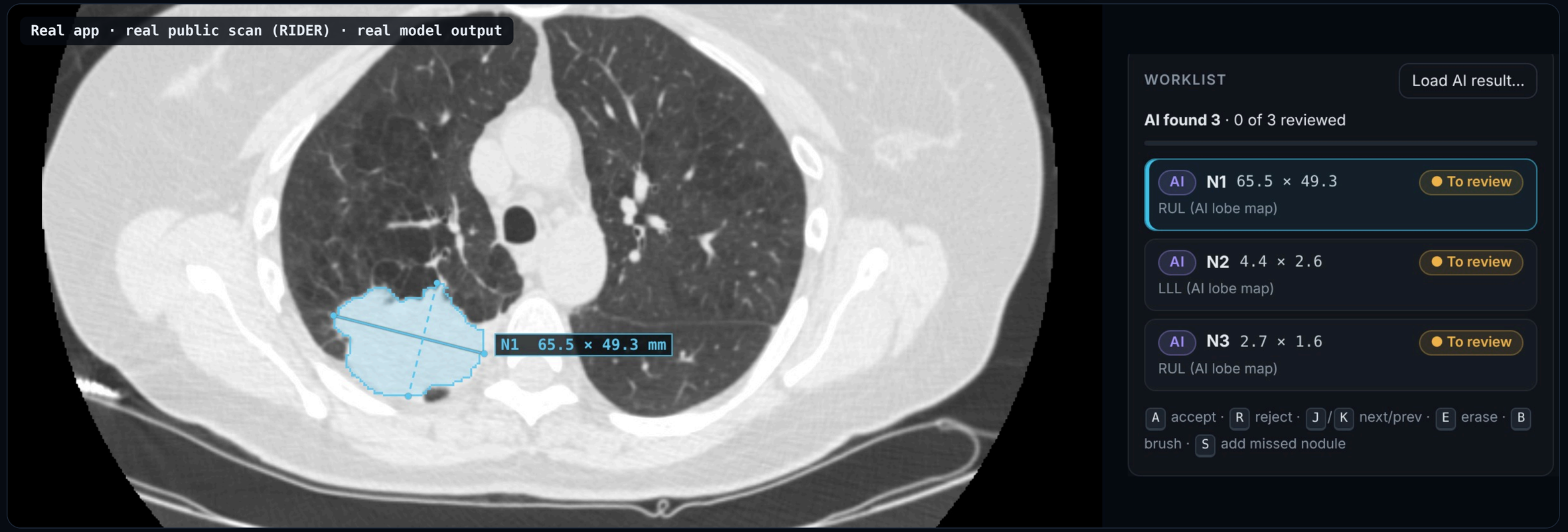
Next (not built yet)

The AI runs by itself on every new export.

AI time measured on 15 LIDC-IDRI scans (MacBook M3 Pro) · privacy checked on the live site

Evidence E6, E22

The AI does the first pass



FINDS

Searches the whole scan

TotalSegmentator, an open AI model, runs on the lab computer. ~97 s per scan.

MEASURES

Outlines and measures

Long and short axis and volume in mm, plus a lobe hint, for every candidate.

WAITS

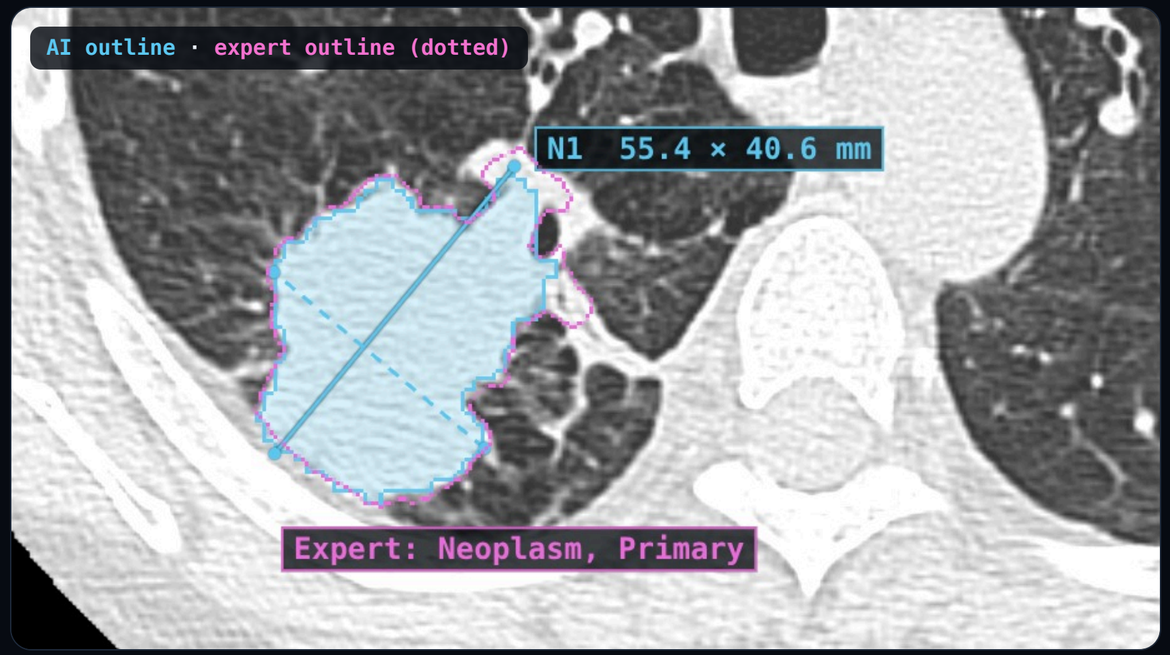
Nothing is final

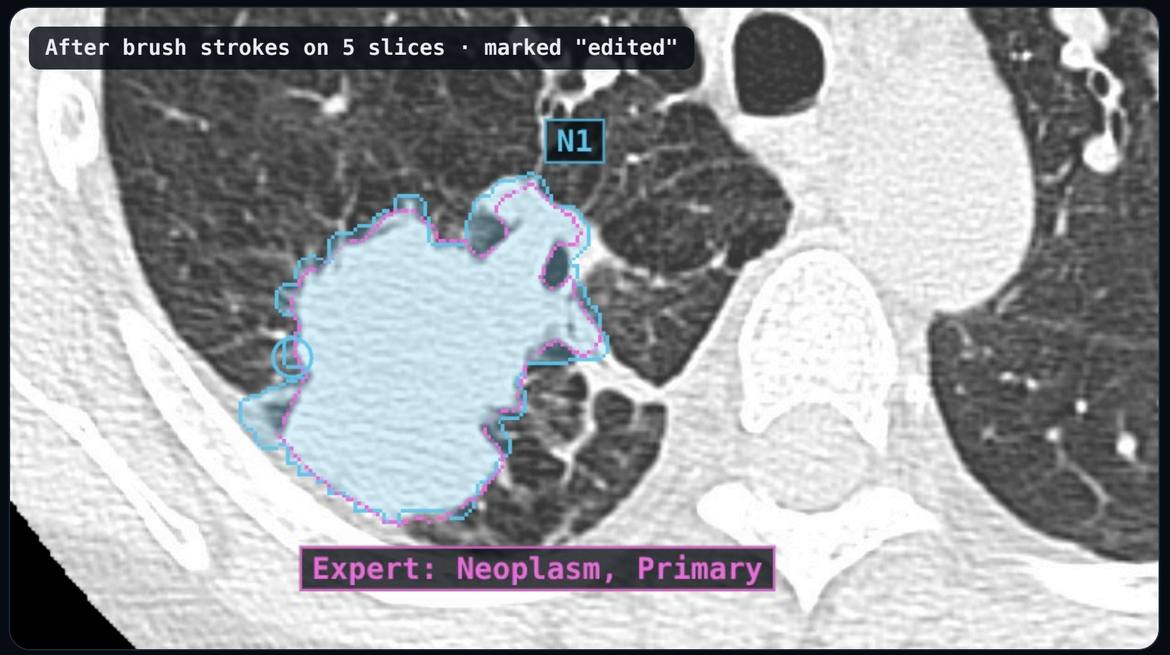
Every candidate starts as "to review". Nothing reaches the sheet until you accept it.

RIDER-1129164940, baseline scan · TotalSegmentator 2.18.0 lung_nodules

Evidence E6, E20

Review. Correct. Approve.





TUMOUR N1	VOLUME	LONG AXIS	DICE
AI proposal	43.8 mL	55.4 mm	0.87
After correction	46.5 mL	62.8 mm	0.88
Expert	55.0 mL	59.8 mm	—

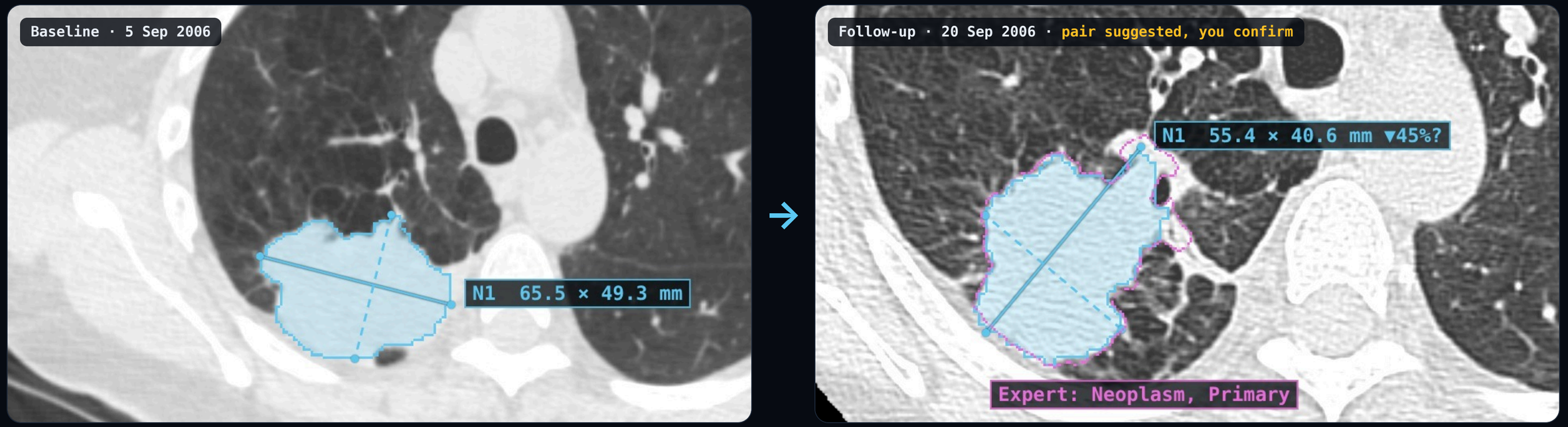
A accept R reject B brush E erase S add missed

All three numbers moved **toward the expert**. Every stroke re-measures, so a wrong stroke shows at once and is one undo away (⌘Z). The strokes here were made by a test script: how long correcting takes for a person is what we still need to measure.

RIDER-1129164940, follow-up scan · expert = RIDER-LungCT-Seg manual outline · values read from the app

Evidence E8, E11

Each nodule gets a suggested match from the earlier scan



LONG AXIS

−15.5 %

65.5 → 55.4 mm

SHORT AXIS

−17.6 %

49.3 → 40.6 mm

VOLUME

−45.1 %

79.7 → 43.8 mL

No more re-finding and side-by-side arithmetic. **Read with care:** the two scans were reconstructed differently, and same-day repeat scans already differ by 10–17 % in volume. Whether it truly shrank is your call.

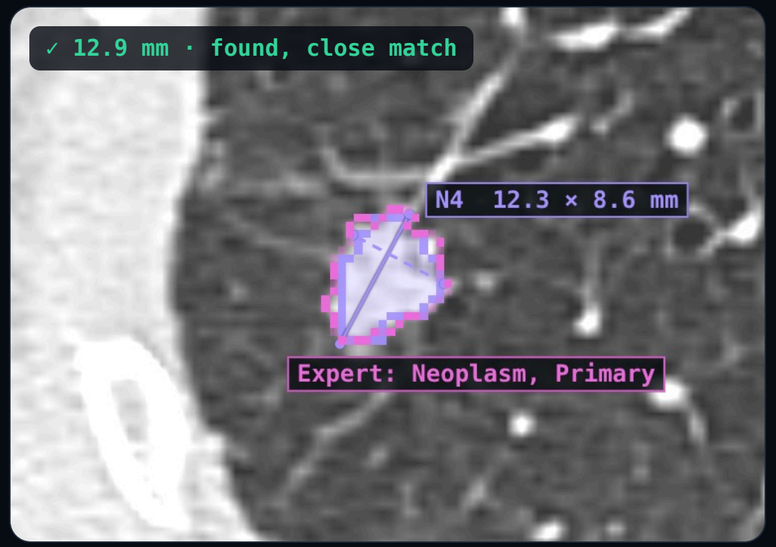
Small solid: found. Ground-glass: not yet.



LIDC · 4 radiologists outlined it (magenta). The AI marked nothing.



RIDER · 13.5 mm tumour, AI outline 8.0 mm. Redraw the edge.



RIDER · AI outline (violet) on the expert's (magenta).

GROUND-GLASS

0 / 5

found

PART-SOLID

1 / 3

found

SOLID

26 / 27

found, but every outline too small

What it means for you: keep searching for hazy nodules yourself. For solid ones, fix the edge before using a volume.

Numbers: LIDC-IDRI, 15 public scans, 35 nodules outlined by ≥ 3 of 4 radiologists, type = their texture rating

Evidence E12–E16, E21

What is proven, and what is not yet

MEASURED ON PUBLIC DATA

53/59 — **scans where the known tumour was found** (RIDER, 31 patients, one labelled tumour per scan)

0.76 — **median overlap with the expert outline** on the 53 found (Dice; 1 = identical)

3/scan — **extra candidates to reject** (median; up to 30 on a few scans)

97 s — **AI time per scan**, unattended, on a laptop

RESEARCHER TIME PER SCAN



> **Not proven yet:** time saved with a real researcher, which will depend mostly on **how long the edge corrections take**; and hazy nodules (ground-glass 0 of 5).

RIDER Lung CT + RIDER-LungCT-Seg, 31 patients · AI time: 15 LIDC scans, MacBook M3 Pro · "today" = Daniel's estimate

Evidence E1–E7, E16, E17

What would make this worth using?

> "... AI-assisted finding followed by my manual review and correction is exactly what I need ... we'll need to time the full process ... and see how it handles those smaller, hazier nodules."

Dr. Daniel Reyes · written feedback

1. On one scan, can you review, correct and export on your own? Where do you get stuck?
2. Given the ground-glass result, would you still use it for solid nodules?
3. How do you get research scans out of the hospital system today?

Next: a 20-minute hands-on session on one scan.

Then a planned timing study: the same task on a few scans, manual vs with the tool, with correction time measured separately.